

## **ANAHAT TRANSFORMATIONS**

Gopal Park, Behind Image Sonography Clinic,  
Paud Phata Rd, Erandwane, Pune 411004

### **Neurological Music Therapy**

Neurological Music Therapy (NMT) is a research-backed therapeutic approach designed to address neurological disorders, including autism. It leverages the inherent connection between music perception and creation and the brain's physiological mechanisms.

Studies indicate that individuals with autism often exhibit a positive response to music, fostering engagement in therapy. Music's universal appeal transcends abilities and disabilities, allowing music therapists to establish rapport from the outset. The adaptability of music, combined with its structured format and sensory input, facilitates a strong connection with the listener.

Autism spectrum disorder is characterized by "qualitative impairments in social interaction and communication" and "restricted repetitive and stereotyped patterns of behavior, interests and activities." Music therapy effectively targets these core characteristics.

By providing sensory stimulation and structure, music therapy can reduce negative behaviors and promote socially appropriate interactions. It can also mitigate self-stimulatory behaviors. For individuals with limited verbal communication, music provides a non-verbal avenue for expression. Furthermore, it can enhance verbal communication, speech, and language skills. Through shared musical experiences, both children and adults with autism can develop skills in shared play, turn-taking, listening, and responding. Music's ability to engage both hemispheres of the brain supports the identification and expression of emotions. It also stimulates cognitive functions and aids in speech and language remediation. Music offers a multi-sensory experience, engaging auditory, visual, and tactile senses.

The rhythmic elements of music influence the sensory system, facilitating control over auditory processing, sensorimotor, and motor skills. Many individuals with autism possess innate musical talents, and music therapy provides a platform for their development. By focusing on internal strengths, therapists can tailor treatment to individual needs.

Advances in brain research, particularly in music cognition and perception, have spurred the development of biomedical applications of music. Music therapy stimulates cognitive, affective, and sensorimotor processes, with applications extending to non-musical therapeutic interventions. Biomedical music research focuses on motor therapy, speech and language therapy, pain management, and cognitive rehabilitation, including memory, executive function, and attention.

Ongoing research is establishing a robust evidence base for NMT. Speech training, leveraging the shared parameters of vocal production between speech and singing, is used to facilitate speech rehabilitation. The use of symbols and signs in both music and language supports language development and the recovery of communication functions.

A structured NMT protocol, delivered by a qualified neurologic music therapist, effectively addresses speech and communication challenges while managing associated neurological issues

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